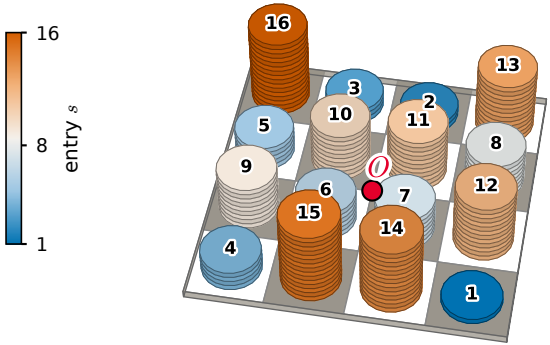
## (a) checkers on a checkerboard

cell $(i, j)$ carries $s_{ij}$ checkers (number and colour = the entry $s_{ij}$)




34 checkers in every row and column;
their centre of mass lies over $O$ ($\sum s\,x = \sum s\,y = 0$)

## (b) the mixed moment $\sum z\,x\,y$

square $S$ (coloured by $z$) and $8\,z\,x\,y$ per cell

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 16 | 3 | 2 | 13 | -135 | +33 | -39 | +81 | Σ 0 |
| 5 | 10 | 11 | 8 | +21 | -3 | +5 | -3 | |
| 9 | 6 | 7 | 12 | +3 | -5 | +3 | -21 | |
| 4 | 15 | 14 | 1 | -81 | +39 | -33 | +135 | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 6 | 11 | 16 | +135 | +15 | +15 | +135 | Σ +480 |
| 8 | 15 | 2 | 9 | +3 | -13 | -13 | +3 | |
| 12 | 3 | 14 | 5 | +21 | -11 | -11 | +21 | |
| 13 | 10 | 7 | 4 | +81 | +9 | +9 | +81 | |